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Midpregnancy maternal circulating angiogenic biomarkers: Associations with placental dysfunction-related pregnancy complications and fetal sex

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A R T I C L E I N F O

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A B S T R A C T

Introduction: Placental function differs by fetal sex, and placental dysfunction is associated with adverse pregnancy outcomes. We aimed to investigate the role of midpregnancy maternal circulating placenta-associated angiogenic biomarkers and fetal sex in relation to placental dysfunction-related pregnancy complications.

Methods: The Preventing Atopic Dermatitis and Allergies in children birth cohort study provided maternal serum from 2511 pregnancies at gestational weeks 16–22. Placental growth factor (PlGF) and soluble fms-like tyrosine kinase-1 (sFlt-1) were analyzed by immunoassays. Pregnancy complications (n = 385) included gestational hypertension, preeclampsia, preterm delivery, and/or newborn weight <10th percentile; categorized as 'uncomplicated', 'complicated' and 'severely complicated' pregnancies by zero, one or two or more complications. The risk of 'any number of pregnancy complications' by biomarkers tertiles were assessed in multivariable logistic regression models with interaction analyses of fetal sex.

Results: Midpregnancy median PlGF was lower and sFlt-1 higher in pregnancies with a female versus a male fetus, with median (interquartile range) sFlt-1/PlGF-ratio being 7.2 (4.9–10.2) versus 6.4 (4.3–9.0) in 'uncomplicated', 7.3 (4.7–10.7) versus 6.4 (4.5–9.0) in 'complicated', and 11.2 (5.4–22.7) versus 5.9 (4.8–13.4) in 'severely complicated' pregnancies. The risks of 'any number of pregnancy complications' were highest in the lowest tertile of PlGF (adjusted odds ratio (aOR) 1.6, 95 % confidence interval (CI): 1.2–2.2) and sFlt-1 (aOR 1.4, 95 % CI: 1.0–1.9) without influence of fetal sex (all, $p_{interactions} > 0.05$).

Conclusion: Circulating midpregnancy placenta-associated angiogenic biomarkers differed by fetal sex and pregnancy complications. The higher risk of pregnancy complications with low midpregnancy PlGF and/or sFlt-1 levels was not influenced by fetal sex.

Abbreviations: Kg/m^2, kilograms per square meters; PlGF, placental growth factor; sFlt-1, soluble fms-like tyrosine kinase-1.

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1. Introduction

Fetal sex differences in placental function have been increasingly acknowledged over the last decade [1,2]. It has been proposed that fetal sex-specific growth pathways exist, suggesting that female fetuses, unlike male fetuses, exhibit less growth without becoming growth-restricted when placental capacity is exceeded [1,2]. This aligns with the generally higher perinatal mortality related to placental dysfunction observed in male fetuses compared to female fetuses [3]. For instance, pregnancies with a male fetus have an overall higher incidence of late-onset, term or post-term preeclampsia [4], gestational hypertension [4], small for gestational age weight percentiles and preterm delivery [3], all representing pregnancy complications associated with a dysfunctional placenta [5].

Maternal circulating levels of proangiogenic placental growth factor (PlGF) and antiangiogenic soluble fms-like tyrosine kinase receptor-1 (sFlt-1), along with the sFlt-1/PlGF-ratio, are recognized as biomarkers of placental function and dysfunction [5]. These angiogenic proteins are produced by the syncytiotrophoblast, the outer cell layer of the placenta villi. They are increasingly used in risk assessments for adverse pregnancy outcomes that are associated with syncytiotrophoblast cellular stress during the second half of pregnancy [5–7]. For instance, for preeclampsia, which becomes clinically overt after 20 weeks of gestation, several prediction cut-offs for the sFlt-1/PlGF ratio valid in the second half of pregnancy, have been proposed [6,8]. However, during the first half of pregnancy, the best documented effective prediction tool for preeclampsia is currently a combination of low maternal PlGF and clinical risk factors [9,10].

Fetal sex-specific differences in maternal circulating PlGF and sFlt-1 during first trimester were first described by Brown et al. [11], and have subsequently been reported by others during the first [12,13], second [12–16], and third trimester [17]. As far as we are aware, none of the existing risk assessment tools for adverse pregnancy outcomes account for the potential influence of fetal sex. Interestingly, in the prediction of early-onset preeclampsia, defined as preeclampsia with delivery before 34 gestational weeks, first trimester PlGF and the sFlt-1/PlGF-ratio

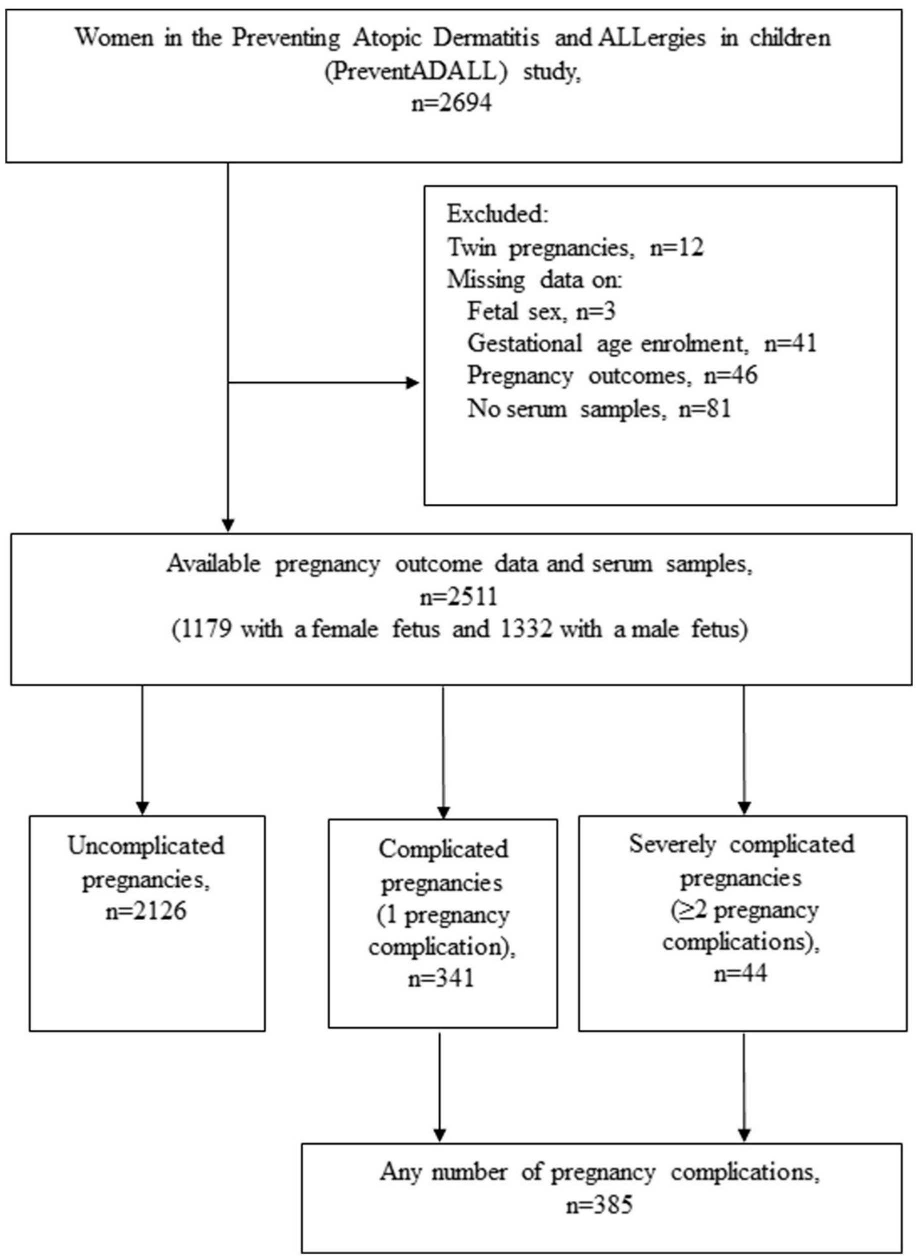

Fig. 1. Flow chart for the study population.

appear to be more informative in pregnancies with a female fetus [12]. Of note, clinically overt early-onset preeclampsia, characterized by very low PlGF and very high sFlt-1 and sFlt-1/PlGF-ratio [7], is the only placental dysfunction-related pregnancy complication that is more prevalent in pregnancies with a female fetus compared to pregnancies with a male fetus [3,4]. Thus, further research is needed to determine whether prediction tools for preeclampsia and other placental dysfunction-related pregnancy complications would benefit from integrating fetal sex into the risk calculator.

We therefore aimed to investigate the role of midpregnancy maternal circulating levels of placenta-associated angiogenic biomarkers (PlGF and sFlt-1) and fetal sex in relation to placental dysfunction-related pregnancy complications.

2. Methods

2.1. Study design and study population

This study included data from the Preventing Atopic Dermatitis and ALLergies in children (PreventADALL) study, a multicenter, prospective mother-child cohort, which has been described in detail previously [18]. In brief, 2694 pregnant women at gestational age 16–22 weeks were recruited from study sites in Norway (Oslo and Østfold) and Sweden (Stockholm) from December 2014 to October 2016. Pregnancies with diagnosed fetal structural anomalies or genetic syndromes were excluded prior to inclusion. Twin pregnancies (n = 12), pregnancies with missing data on fetal sex (n = 3), gestational age at enrolment (n = 41) or pregnancy outcome (n = 46) and pregnancies with unavailable serum sample for biomarker analysis (n = 81) were excluded in the present study, yielding 2511 singleton pregnancies available for analysis (Fig. 1). Participant data were retrieved from comprehensive electronic questionnaires at gestational age 18 and 34 weeks, medical records and national birth registries.

The general population-based design of the PreventADALL study allows us to investigate biomarkers within a representative group that reflects common underlying conditions. Pregestational and gestational diabetes mellitus as well as chronic hypertension, along with other maternal chronic conditions, were not excluded from our cohort, although maternal diabetes and chronic hypertension specifically are associated with an increased risk of placental dysfunction [19]. Gestational diabetes mellitus was diagnosed according to Norwegian national guidelines, defined by a 2-h plasma glucose level of ≥7.8 mmol/L and <11.1 mmol/L [20]. Chronic hypertension was diagnosed per ISSHP 2014 criteria, defined as a systolic blood pressure ≥140 mmHg and/or diastolic blood pressure ≥90 mmHg prior to 20 weeks' gestation [21].

2.2. Ethics

Written informed consent was obtained upon enrolment. The study was approved by the Regional Committee for Medical and Health Research Ethics in South-Eastern Norway (2014/518; December 8th, 2014) and the Ethics committee in Sweden (2014/2242–31/4) prior to study inclusion. The PreventADALL study is registered in ClinicalTrials.gov (reference number NCT02449850).

2.3. Blood sampling and biomarker analysis

Non-fasting maternal blood was drawn at inclusion and serum stored at minus 80°C until analyses were conducted in 2018. After thawing, serum was analyzed for the placenta-associated angiogenic biomarkers sFlt-1 and PlGF concentrations at one site (Oslo University Hospital, Department of Medical Biochemistry), using the fully automated Elecsys® PlGF and sFlt-1 assays on the Cobas e801 electrochemiluminescence immunoassay platform (Roche Diagnostics) according to the manufacturer's instructions. All concentrations were within the detectable ranges of the PlGF and sFlt-1 assays (3–10000 pg per milliliters (pg/mL) and 10–85000 pg/mL, respectively). The coefficients of variation were ≤1.8 % for sFlt-1 and ≤2.1 % for PlGF. All clinicians were blinded to biomarker data and all laboratory personnel blinded to clinical data.

2.4. Outcome variables

Data on pregnancy characteristics and pregnancy outcomes were recorded in the patient chart and the hospital delivery register (PARTUS), which subsequently provided the bases for study personnel recording delivery and birth data in the PreventADALL study. The following placental dysfunction-related pregnancy complications outcomes of interest were included in the present study (see Table 1): gestational hypertension, preeclampsia, small for gestational age weight percentiles and preterm delivery of any cause. Preeclampsia, gestational hypertension and chronic hypertension diagnoses were defined according to the 2014 International Society for the Study of Hypertension in pregnancy (ISSHP) criteria [21]. Small for gestational age weight ≤10th percentile was defined according to gestational age and fetal sex, and was based on Norwegian reference ranges [22]. Preterm delivery was defined as birth prior to gestational age 37 weeks [23]. The main placental dysfunction-related pregnancy complications outcomes were defined as (Fig. 1) 'uncomplicated pregnancy' including normotensive women who delivered newborns with normal fetal weight percentiles at term, 'complicated pregnancy' with one complication, 'severely complicated pregnancy' with two or more complications. Furthermore, for the regression models we used uncomplicated pregnancy as the reference versus 'any number of pregnancy complications'.

2.5. Exposure variable

The main exposures were maternal serum placenta-associated angiogenic biomarkers PlGF and sFlt-1 in pg/mL, and the sFlt-1/PlGF-ratio. For the regression analyses, the biomarkers were categorized into tertiles to distinguish between high and low levels, and to capture the skewed distribution. Based on previous literature, we chose to

Table 1
Definitions of placenta dysfunction-related pregnancy complications used in the present study.

Pregnancy complication	Definition	Reference
Gestational hypertension	New onset hypertension (systolic blood pressure ≥140 mmHg and/or diastolic blood pressure ≥90 mmHg) after gestational age 20 weeks in previously normotensive women in the absence of proteinuria and other preeclampsia-associated signs	[33]
Preeclampsia	New onset gestational hypertension and new proteinuria, defined as either protein dip stick ≥1+ present on last documented urine testing or a total protein (milligrams/liter): creatinine (millimoles/liter) ratio ≥30, or a 24-h urine excretion of ≥0.3 g protein, in the absence of urinary infection after gestational age 20 weeks in previously normotensive women.	[33]
Early-onset preeclampsia	Preeclampsia with a delivery < gestational age 34 weeks	[33]
Late-onset preeclampsia	Preeclampsia with a delivery ≥ gestational age 34 weeks	[33]
Superimposed preeclampsia	Chronic hypertension < gestational age 20 weeks with the addition of proteinuria ≥ gestational age 20 weeks	[33]
Small for gestational age weight percentiles	Newborn birth weight below 10th percentile for gestational age at birth	[34]
Preterm delivery	Delivery (of any cause) < gestational age 37 weeks	[35]

compare the middle and lowest tertiles to the highest tertile of PlGF [7,24], and the highest and lowest tertiles to the middle tertile of sFlt-1 [7,24,25]. Tertile thresholds were calculated separately for two gestational age periods '16–18 weeks' and '19–22 weeks' (Supplementary Table 1). Gestational age in weeks was derived by fetal femur length at the routine second trimester ultrasound examination.

2.6. Covariates

Covariates included known risk factors for preeclampsia [26]: maternal prepregnancy body mass index in kg/m^2 (kilograms per square meters) based on weight and height measured at inclusion (dichotomized into <28 vs ≥ 28 kg/m^2), maternal age at enrolment (dichotomized into ≤40 vs > 40 years), nulliparity (0, >1) and fetal sex assigned at birth (female, male). Gestational age was not included as a covariate, as the tertile thresholds were calculated for gestational age periods prior to deriving the exposure variables.

2.7. Statistical analyses

For descriptive statistics, categorical variables are presented as frequencies with percentages (%) and continuous variables as means with standard deviations (SDs) or medians with interquartile ranges (IQR). Participant characteristics are presented by adverse pregnancy outcome group.

We calculated percentiles (5th, 33rd, 50th, 67th, 95th) for concentrations of PlGF, sFlt-1 and the sFlt-1/PlGF- ratio in the uncomplicated pregnancy group (n = 2126) by fetal sex. The percentiles were calculated by gestational week intervals '16–17', '18', '19' and '20–22′, due to the rapidly increasing PlGF values in this gestational age period [7].

To investigate associations between midpregnancy levels for the placenta-associated biomarkers PlGF, sFlt-1 and the sFlt-1/PlGF-ratio and pregnancy complications, we describe median (IQR) biomarker levels by adverse pregnancy outcome group and fetal sex. We further used univariable and multivariable logistic regression models with biomarker tertiles as exposure and 'any number of pregnancy complications' as outcome. The highest PlGF tertile, the middle sFlt-1 tertile and the lowest sFlt-1/PlGF tertile were set as reference categories based on tertile thresholds, as detailed in Supplementary Table 1 [7]. A Directed Acyclic Graph was used to select the minimal set of covariates necessary to estimate the total association between exposures and outcomes in our models (Supplementary Fig. 1) [27]. The adjusted models included the covariates prepregnancy body mass index, maternal age, nulliparity and fetal sex. A total of 11.6 % of the data were missing for the covariates (maternal prepregnancy body mass index; n = 288, parity; n = 1, sBP and dBP pre-20 weeks' gestation; n = 24, highest sBP last two weeks of pregnancy; n = 604) and were excluded from all regression models (complete case analysis).

To investigate if the associations between biomarker levels and pregnancy complications differed by fetal sex, an interaction term

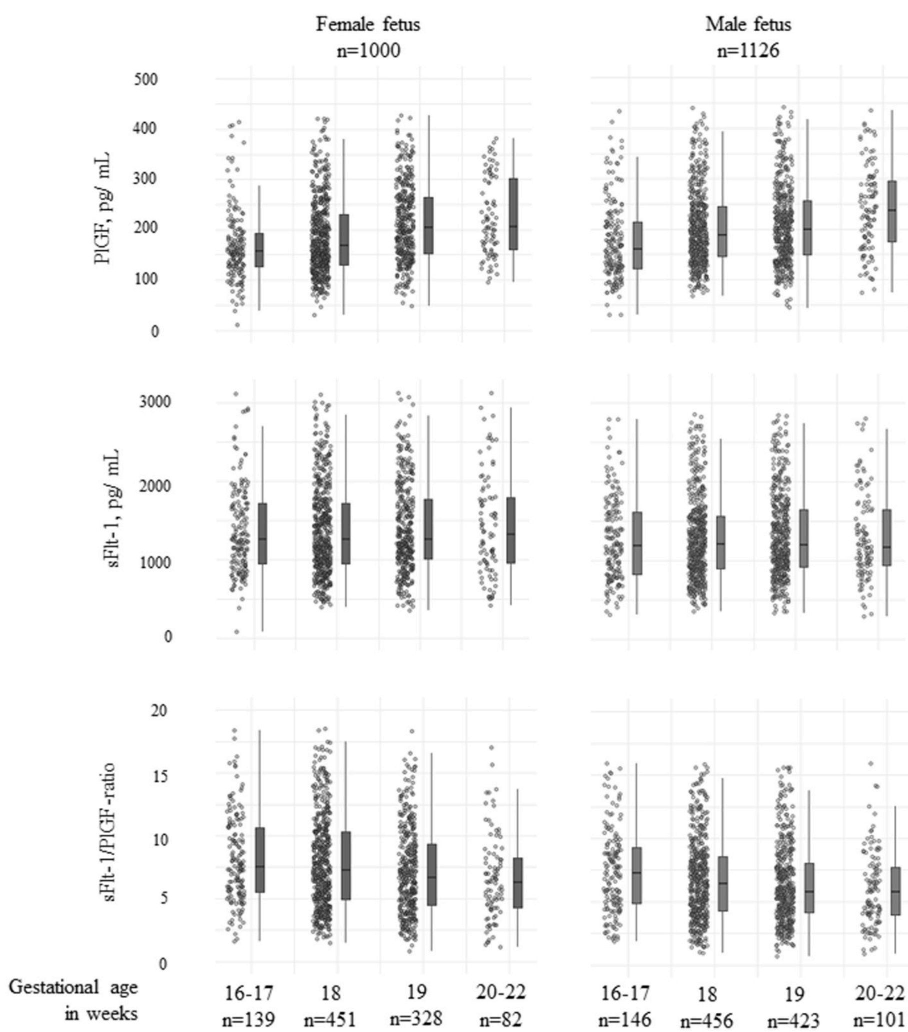

Fig. 2. Normal group percentiles: maternal levels of midpregnancy PlGF, sFlt-1 and sFlt-1/PlGF-ratio in the 'uncomplicated pregnancy' group, by fetal sex Abbreviations: n = number; pg/mL = picograms per milliliters; PlGF = placental growth factor; sFlt-1 = soluble fms-like tyrosine kinase-1.

between fetal sex and biomarker tertile was added in the adjusted models. In case of significant interaction, the results of the logistic regression are presented stratified for female and male fetal sex.

All analyses were performed by IBM SPSS Statistics version 30 (Chicago, IL, the USA). The data visualization presented in Figs. 2 and 3 was performed in R (version 4.4.2) using the ggplot package. A p-value <0.05 was considered statistically significant.

3. Results

The mean maternal age at enrolment was 32 years, and mean gestational age at delivery was 40 weeks (Table 2). The time from inclusion to delivery was 21.0 weeks for the total cohort (2511 singleton pregnancies, data not shown). In total 2126 (84.7 %) pregnancies were classified as 'uncomplicated pregnancy' and 385 (15.3 %) pregnancies had at least one placental dysfunction-related pregnancy complication: gestational hypertension (4 %), preeclampsia (2.1 %), small for gestational age newborn weight ≤10th percentile (7 %) and preterm delivery of any cause (4 %). Overall, 341 (13.6 %) were categorized as 'complicated pregnancies' and 44 (1.8 %) as 'severely complicated pregnancies' (Table 2, Fig. 1). In the 'complicated pregnancy' group, 2 % of the study participants had chronic hypertension and 7 % gestational diabetes mellitus. In the 'severely complicated group', these conditions were present in 2 % and 5 % of the participants, respectively (Table 2). Pregnancies with a female fetus less often developed 'severely complicated pregnancy' compared to those of male fetus (41 % versus 59 %, Table 2).

3.1. Midpregnancy PlGF, sFlt-1 and sFlt1-/PlGF in the uncomplicated pregnancy group

For the 'uncomplicated pregnancy' group, each biomarker is presented by gestational age weeks intervals for pregnancies with a female fetus and a male fetus in Fig. 2 and in Supplementary Table 2. The median PlGF level was higher for the more advanced gestational age groups, regardless of fetal sex, whereas median sFlt-1 was relatively stable across the groups. Consequently, median sFlt-1/PlGF-ratio was slightly lower in the more advanced gestational age groups. Additionally, in this group of 'uncomplicated pregnancies', the pregnancy group with a female fetus had a lower median PlGF level and higher median sFlt-1 and sFlt-1/PlGF-ratio compared to pregnancies with a male fetus (Fig. 2, Supplementary Table 2).

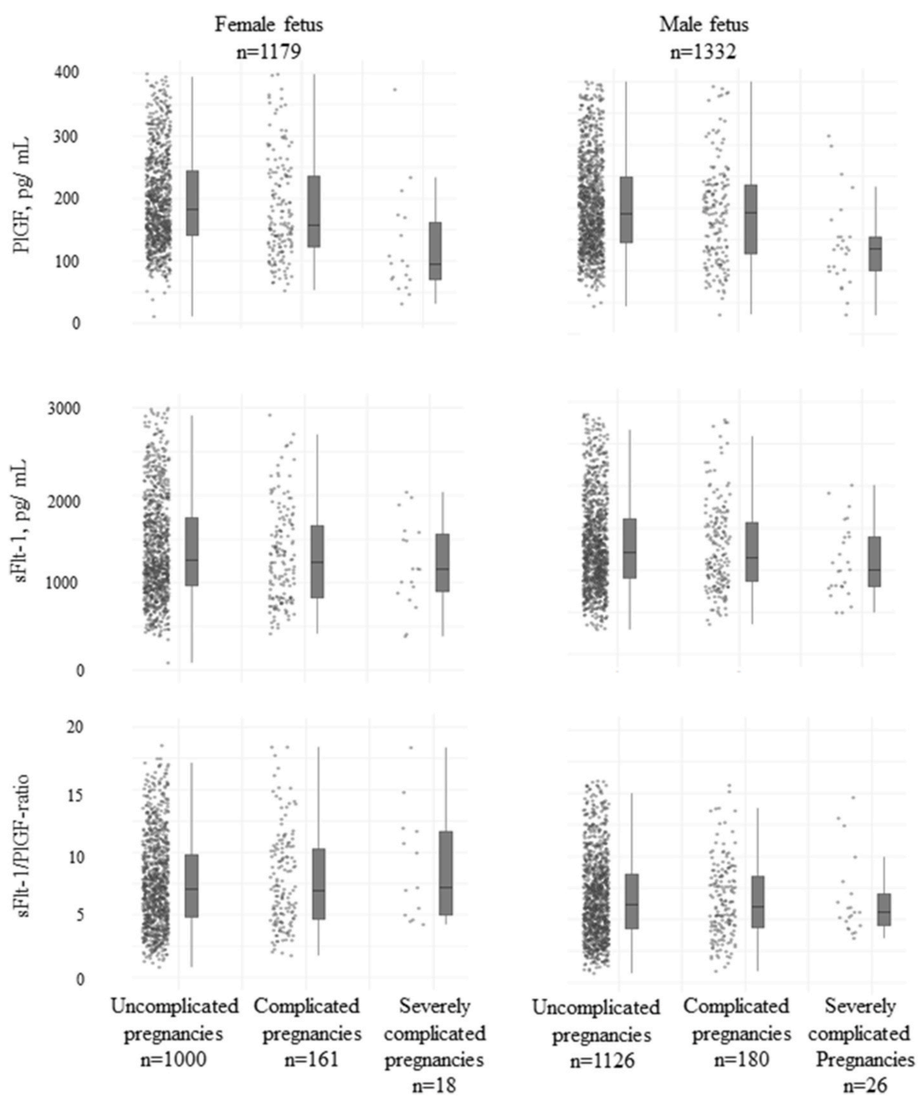

Fig. 3. Maternal levels of midpregnancy PlGF, sFlt-1 and sFlt-1/PlGF-ratio across the study groups, by fetal sex. 'Complicated pregnancy' group: one (1) pregnancy complication. 'Severely complicated pregnancy' group: two or more (≥2) pregnancy complications. Abbreviations: n = number; pg/mL = picograms per milliliters; PlGF = placental growth factor; sFlt-1 = soluble fms-like tyrosine kinase-1.

Table 2
Clinical and midpregnancy biomarker descriptives across the study groups. Numbers are given as n (%) or mean ± standard deviation and biomarker levels as median (interquartile range).

	Total	Uncomplicated pregnancies	Complicated pregnancies (1 pregnancy complication)	Severely complicated pregnancies (≥2 pregnancy complications)
	n = 2511 (100 %)	n = 2126 (84.7 %)	n = 341 (13.6 %)	n = 44 (1.8 %)
Female fetal sex	1179 (47 %)	1000 (47 %)	161 (47 %)	18 (41 %)
Nulliparous	1490 (59 %)	1208 (57 %)	243 (71 %)	39 (89 %)
Gestational age at study inclusion	18.9 ± 0.9	18.9 ± 0.9	18.9 ± 0.9	18.8 ± 0.7
Maternal age at enrolment	32.3 ± 4.2	32.3 ± 4.2	32.2 ± 4.6	32.4 ± 4.5
Maternal age >40 years	79 (3 %)	58 (3 %)	18 (5 %)	3 (7 %)
[a]Maternal prepregnancy body mass index in kg/m^2	24.8 ± 3.7	24.7 ± 3.5	25.2 ± 4.5	26.4 ± 4.7
[a]Maternal body mass index ≥28 kg/m^2	352 (14 %)	290 (14 %)	54 (16 %)	8 (18 %)
[a]sBP pre-20 weeks' gestation	108.8 ± 10.1	108.4 ± 9.8	111.3 ± 11.1	111.4 ± 10.5
[a]dBP pre-20 weeks' gestation	61.7 ± 7.0	61.4 ± 6.7	63.7 ± 7.9	63.6 ± 8.4
Chronic hypertension	28 (1 %)	22 (1 %)	5 (2 %)	1 (2 %)
Gestational diabetes mellitus	92 (4 %)	67 (3 %)	23 (7 %)	2 (5 %)
Gestational age at birth in weeks	39.9 ± 1.8	40.2 ± 1.2	38.7 ± 3.0	36.2 ± 3.0
[a]Highest sBP last two weeks of pregnancy in mmHg	120.1 ± 13.5	117.9 ± 10.8	129.6 ± 17.1	145.0 ± 26.7
[a]Highest dBP last two weeks of pregnancy in mmHg	75.9 ± 9.7	74.4 ± 8.3	82.8 ± 11.6	91.3 ± 13.5
Gestational hypertension	92 (4 %)		80 (23 %)	12 (27 %)
Late-onset preeclampsia	60 (2 %)		43 (12 %)	17 (39 %)
Early-onset preeclampsia	3 (0.1 %)		0 (0 %)	3 (7 %)
Small for gestational age weight percentiles	186 (7 %)		151 (43 %)	35 (80 %)
Preterm delivery of any cause <37 weeks	101 (4 %)		73 (21 %)	28 (64 %)
PlGF, pg/mL	192 (142–261)	194 (147–264)	181 (127–245)	118 (75–165)
sFlt-1, pg/mL	1255 (937–1752)	1265 (951–1777)	1231 (841–1668)	1047 (805–1489)
sFlt-1/PlGF-ratio	6.8 (4.5–9.7)	6.8 (4.5–9.6)	6.9 (4.6–9.9)	7.1 (5.0–17.5)

Abbreviations: dBP = diastolic blood pressure; kg/m^2 = kilograms per square meter; n = number; % = percentage; pg/mL = picograms per milliliters; PlGF = placental growth factor; sBP = systolic blood pressure; sFlt-1 = soluble fms-like tyrosine kinase-1.
[a]Missing data: maternal body mass index; n = 288, sBP and dBP pre-20 weeks' gestation; n = 24, highest sBP last two weeks of pregnancy; n = 604.

3.2. *Midpregnancy PlGF, sFlt-1 and sFlt1-/PlGF and pregnancy complications*

Compared to 'uncomplicated pregnancy', we observed lower median midpregnancy levels of PlGF and sFlt-1 in the 'complicated pregnancy' group, and even lower levels in the 'severely complicated pregnancy' group (Table 2). Compared to pregnancies with a male fetus, pregnancies with a female fetus exhibited a lower median maternal level of PlGF at midpregnancy, as well as higher median sFlt-1 level and sFlt-1/PlGF-ratio (Table 3, Fig. 3). In 'severely complicated pregnancy' the median maternal level of the sFlt-1/PlGF-ratio was particularly high in pregnancies with a female compared to a male fetus (11.2 versus 5.9, Table 3).

The risk of 'any pregnancy complications' increased with lower midpregnancy PlGF levels, with the highest odds among pregnancies with the lowest compared to highest tertile (adjusted odds ratio (aOR) 1.6, 95 % confidence interval (CI): 1.2–2.2) (Table 4) ($p = 0.008$). A similar dose-response association was observed for the sFlt-1 level with lower odds of 'any pregnancy complications' in the highest tertile (aOR 0.8, 95 % CI 0.6–1.1) and higher odds in the lowest tertile (aOR 1.4, 95 % CI: 1.0–1.9) compared to the middle (reference) tertile ($p < 0.001$; Table 4). No association was found between the midpregnancy sFlt-1/PlGF-ratio and odds for developing 'any number of pregnancy complications' ($p = 0.363$). The associations between midpregnancy biomarkers and 'any number of pregnancy complications' did not differ by fetal sex (all $p_{interaction} > 0.05$; Table 4).

4. Discussion

Maternal circulating levels of placenta-associated angiogenic biomarkers PlGF and sFlt-1 were observed to have a more antiangiogenic (i.e. a less proangiogenic) biomarker profile (lower PlGF level and higher sFlt-1 and sFlt-1/PlGF-ratio) at midpregnancy in pregnancies with a female fetus compared to those with a male fetus, across all pregnancy outcome groups. However, fetal sex did not affect the association between low maternal levels of PlGF and/or sFlt-1 and the odds of subsequently developing any number of pregnancy complication.

Our finding of lower circulating midpregnancy PlGF levels in uncomplicated pregnancies with a female fetus compared to those with a male fetus is in line with a previous very small study (n = 38; 19 males and 19 females) [28]. This study found lower PlGF serum levels longitudinally in uncomplicated pregnancies with a female fetus compared to those with a male fetus [28]. Furthermore, a previous report showed that PlGF was higher during first trimester, and lower during second trimester in pregnancies with a female fetus compared to those with a male fetus [13]. Further, we found a slightly higher midpregnancy sFlt-1 in uncomplicated pregnancies with a female fetus compared to a male fetus. In contrast to our findings, a previous study reported that fetal sex-specific differences were not present in maternal sFlt-1 levels in normotensive second-trimester pregnancies (n = 328, gestational age ranging from 9 to 35 weeks) [15]. We speculate that lower levels of PlGF and sFlt-1 in uncomplicated pregnancies with a female fetus may correlate with smaller placentae, as evidenced by reduced placental weight and lower birth weight-to-placental weight ratios in our study

Table 3
Maternal circulating placenta-associated angiogenic biomarkers given by fetal sex across the study groups. Midpregnancy angiogenic biomarker levels are given in median (interquartile range). 'Complicated pregnancy' group: one (1) pregnancy complication. 'Severely complicated pregnancy' group: two or more (≥2) pregnancy complications.

Pregnancies with a female fetus n = 1179			
	Uncomplicated pregnancy n = 1000 (84.8 %)	Complicated pregnancy (1 pregnancy complication) n = 161 (13.7 %)	Severely complicated pregnancy (≥2 pregnancy complications) n = 18 (1.5 %)
PlGF, pg/mL	189 (143–259)	160 (123–243)	94 (67–170)
sFlt-1, pg/mL	1301 (989–1857)	1266 (829–1728)	1155 (862–1582)
sFlt-1/PlGF-ratio	7.2 (4.9–10.2)	7.3 (4.7–10.7)	11.2 (5.4–22.7)
Pregnancies with a male fetus n = 1332			
	Uncomplicated pregnancy n = 1126 (84.5 %)	Complicated pregnancy (1 pregnancy complication) n = 180 (13.5 %)	Severely complicated pregnancy (≥2 pregnancy complications) n = 26 (2 %)
PlGF, pg/mL	200 (149–268)	197 (132–247)	135 (99–161)
sFlt-1, pg/mL	1235 (920–1707)	1168 (875–1617)	1000 (799–1407)
sFlt-1/PlGF-ratio	6.4 (4.3–9.0)	6.4 (4.5–9.0)	5.9 (4.8–13.4)

Abbreviations: n = number; % = percentage; pg/mL = picograms per milliliters; PlGF = placental growth factor; sFlt-1 = soluble fms-like tyrosine kinase-1.

(data not shown) and prior research [29–31]. Additionally, we speculate that these maternal angiogenic biomarker differences at midpregnancy may be the result of a differing placentation process across fetal sex and thereby differing placental function and circulating maternal PlGF and sFlt-1.

Previous studies have shown sex differences in cytotrophoblast and syncytiotrophoblast gene expression [32], likely influenced by sex chromosomes during early placentation [33]. For instance, sexual dimorphism in the gene expression of placental villous tissue has been associated with VEGF during both the second trimester and at term. Given that PlGF is a member of the VEGF family [34,35], and these proteins have crucial roles in trophoblast invasion during placentation [36,37], we suggest that the variations in PlGF levels observed in our study may reflect sex-dependent differences in implantation and placentation.

The pattern of lower midpregnancy PlGF was also present in the complicated and severely complicated pregnancy groups, regardless of fetal sex (Table 3 and Fig. 3). Our findings are in line with a "dose-response" effect, showing more dysregulated angiogenic biomarker levels in women with a higher number of placental dysfunction-related pregnancy complications. We also show in the 'severely complicated pregnancy' group that maternal levels of PlGF were notably low and the sFlt-1/PlGF-ratio particularly high in pregnancies with a female compared to a male fetus. This sexual dimorphism highlights a nuanced interaction between underlying biological factors and emphasizes the importance of considering fetal sex when evaluating these biomarkers. In agreement with our findings, a similar trend has previously been reported, in that more precise preeclampsia prediction using PlGF and the sFlt-1/PlGF-ratio has been seen in pregnancies with a female compared to a male fetus. One such study had longitudinal angiogenic biomarker samples (PlGF and sFlt-1/PlGF-ratio) at gestational weeks 8–14 (n = 2110) and 20 to 34 (n = 1595, of these 112 with preeclampsia) [12]. Another study showed the same trend for sFlt-1, with samples from gestational weeks 9–25 (n = 573, of these 300 with preeclampsia) [15]. Additionally, while evidence regarding morphological sex differences in the human placentation process remains limited, some studies suggest sex dimorphism in morphological features. Evidence indicates that villous trophoblast cells (cytotrophoblasts and syncytiotrophoblast) exhibit greater proliferative activity in healthy male placentas compared to female placentas [38], whereas female preeclamptic placentas show greater proliferative activity than those from males [38,39]. This supports the notion that females may be more adaptive to the intrauterine environment than males [2]. These sex differences in placental cell function align with our findings of significantly elevated sFlt-1 levels and a higher sFlt-1/PlGF ratio in severely complicated pregnancies with female fetuses.

Our observed association between low midpregnancy maternal level of PlGF and sFlt-1 and higher odds of developing placental dysfunction-related pregnancy complications corresponds with earlier reports. A previous study comprised 30 013 pregnancies, thereof 1501 with a PlGF level below the 5th percentile at gestational age 19–23 weeks. The latter group had higher rates, compared to the pregnancies with PLGF levels above the 5th percentile, of subsequent preeclampsia, gestational hypertension, fetal growth restriction, preterm birth and stillbirth [40].

Table 4
Association between maternal levels of placenta-associated angiogenic biomarkers by tertiles with "any number of pregnancy complications" (1 or more pregnancy complications) from univariable and multivariable logistic regression models, n = 2223 (complete case analysis). Tertiles thresholds by gestational age are specified in Supplementary Table 1. The following covariates were included in the adjusted analysis: maternal prepregnancy body mass index in kg/m2 (<28, ≥28), maternal age in years at enrolment (≤40, >40), parity (0, >1), fetal sex (male, female), as specified in Supplementary Fig. 1 $p_{interaction}$ denotes the interaction between angiogenic biomarker by tertiles and fetal sex.

	"Any number of pregnancy complications", n	Unadjusted OR (95 % CI)	P-value	Adjusted OR (95 % CI)	P-value	$P_{interaction}$
PlGF, pg/mL			**0.005**		**0.008**	0.715
Lowest tertile	123	1.6 (1.2–2.2)		1.6 (1.2–2.2)		
Middle tertile	94	1.2 (0.9–1.7)		1.2 (0.9–1.7)		
Highest tertile, ref.	80	1.0		1.0		
sFlt-1, pg/mL			**0.006**		**<0.001**	0.899
Lowest tertile	122	1.3 (1.0–1.7)		1.4 (1.0–1.9)		
Middle tertile, ref.	94	1.0		1.0		
Highest tertile	81	1.0 (0.7–1.3)		0.8 (0.6–1.1)		
sFlt-1/PlGF-ratio			0.165		0.363	0.203
Lowest tertile, ref.	79	1.0		1.0		
Middle tertile	106	0.9 (0.6–1.2)		0.9 (0.6–1.2)		
Highest tertile	112	1.2 (0.8–1.6)		1.0 (0.8–1.4)		

Abbreviations: CI = confidence interval; n = number; OR = odds ratio; pg/mL = picograms per milliliter; PlGF = placental growth factor; P-value = global p value; $P_{interaction}$ = p-value for interaction; ref. = reference group; sFlt-1 = soluble fms-like tyrosine kinase-1.

Furthermore, low PlGF is found in the first trimester [41], as well as at midpregnancy during weeks 22–34 [42], in women who subsequently develop spontaneous preterm birth. Another study (n = 801) reported low levels of sFlt-1 at gestational age 15–22 weeks in pregnancies that subsequently had a medically indicated preterm delivery for a reason independent of preeclampsia or small for gestational age fetus (n = 71) [25]. Additionally, low first trimester PlGF and sFlt-1 levels are also previously reported in pregnancies with subsequent composite placental dysfunction-related pregnancy outcomes (n = 418, of these 208 cases) [24]. In contrast, one study (n = 600, of these 25 with preeclampsia) reported that maternal levels of midpregnancy sFlt-1 did not differ between women developing preterm preeclampsia (delivery <37 weeks), late-onset preeclampsia (delivery <34 weeks) and uncomplicated pregnancies [9]. Based on our general population-based study design, which aims to represent the broader pregnant population and supported by previous reports [9,25,40–42], we propose that midpregnancy maternal levels of PlGF and sFlt-1 serve as general biomarkers of placental health. These could be used in identifying a wide range of women with increased risk for subsequent placental dysfunction-related complications, with low levels at midpregnancy indicating a higher risk for adverse pregnancy outcomes.

The sFlt-1/PlGF-ratio at midpregnancy was not associated with increased odds of placental dysfunction-related pregnancy complications in our study. Several thresholds for the sFlt-1/PlGF-ratio have been suggested for gestational weeks 20 [43], or 24 to 37 [6,8], and after 37 weeks [44], in predicting future absence or presence of preeclampsia, preterm delivery and/or uteroplacental dysfunction, such as a ratio ≥38 and/or ≥85. In our clinically healthy, general population-based cohort, only five pregnancies had a sFlt-1/PlGF-ratio ≥38 (data not shown). Supported by previous literature, we propose that PlGF and/or sFlt-1 may prove more useful than their ratio as risk prediction markers during the first half of pregnancy for subsequent placenta-related adverse pregnancy outcomes.

As the odds of developing any number of pregnancy complications in the presence of low midpregnancy PlGF and sFlt-1 levels were not influenced by fetal sex in our study, we speculate that midpregnancy prediction tools for adverse pregnancy outcomes associated with placental dysfunction are unlikely to benefit from the addition of fetal sex. On the other hand, we cannot rule out that fetal sex differences in placenta-associated angiogenic biomarkers might influence risk stratification during the latter half of pregnancy, when complications related to placental dysfunction, such as preeclampsia, become clinically overt. Our interpretation of our data is that the midpregnancy differences in biomarker levels illustrate a biological variation in placental function between female and male fetus pregnancies.

5. Strengths and limitations

A major strength of the present study is the large number of study participants with available serum samples within a relatively small gestational age window at midpregnancy, which enables prospective study design. We present a well-described homogenous pregnancy population included at three study sites, with a relatively similar incidence of placental dysfunction as previously reported (15.3 % versus 10–15 %) [45]. Another strength of our study is that we retained the whole general population-based cohort in our analysis, for instance women with pregestational diabetes mellitus, chronic hypertension, gestational diabetes mellitus, and preterm birth (from any cause). Excluding any such groups would undermine the prospective, population-based design of the PreventADALL cohort and reduce our capacity to identify subsequent clinical manifestations of potential placental dysfunction within a representative cohort reflecting the general population. Placental dysfunction, often secondary to shallow placentation with uteroplacental vascular lesions, is a contributor to spontaneous and medically indicated preterm births alongside other factors such as infection, inflammation and cervical dysfunction and injury [46]. Furthermore, preexisting maternal conditions related to endothelial dysfunction, such as pregestational diabetes mellitus, may lower the threshold for excessive maternal vascular inflammation and accelerate syncytiotrophoblast senescence, thereby increasing the risk of preeclampsia development [19]. During the inclusion at 16–22 weeks' of gestation, the future potential cases of gestational diabetes mellitus were mostly undiagnosed in our cohort, as the oral glucose tolerance test was typically performed around 28 weeks' gestation [20]. Excluding specific pregnancy outcome groups such as gestational diabetes mellitus, chronic hypertension or preterm delivery of a non-placental cause could introduce selection bias and limit the generalizability of our midpregnancy biomarker findings.

Study limitations include low incidences of severe placental dysfunction-related pregnancy complications. For instance, our cohort had a lower rate of preterm delivery (4 % versus 6.4 %) and a lower incidence of preeclampsia (2.1 % compared to 2.7 %) compared to the general Norwegian population and to global preeclampsia rates (3–10 %) [47–50]. Furthermore, the rates of gestational hypertension and small for gestational age were lower in our cohort compared to the Medical Birth Registry of Norway (4 % compared to 1.6 % and 7 % compared to 4.8 %, respectively) [47,51]. In addition, the PreventADALL cohort has low ethnic diversity, and consists of predominantly highly educated women [18,52], partly because of inclusion criteria (fluency in a Scandinavian language) [18].

6. Conclusion

The midpregnancy placenta-associated angiogenic biomarkers PlGF and sFlt-1 were associated with placental dysfunction-related pregnancy complications, with lower PlGF and higher sFlt-1 in pregnancies with a female fetus compared to a male fetus. The risk of placental dysfunction-related pregnancy complications in pregnancies was higher with low midpregnancy maternal levels of PlGF and sFlt-1, with no fetal sex interaction. Consequently, we suggest that midpregnancy risk calculators for placental dysfunction-related pregnancy outcomes, that already include angiogenic protein levels, are unlikely to be improved by adding fetal sex.

CRediT authorship contribution statement

Birgitte Kordt Sundet: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Karin C. Lødrup Carlsen:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Guttorm Haugen:** Writing – review & editing, Project administration. **Gunilla Hedlin:** Writing – review & editing, Project administration. **Katarina Hilde:** Data curation, Writing – Review & editing.**Christine M. Jonassen:** Writing – review & editing, Project administration. **Björn Nordlund:** Writing – review & editing, Project administration. **Eva Maria Rehbinder:** Writing – review & editing, Project administration. **Corina Silvia Rueegg:** Writing – review & editing, Visualization, Validation, Methodology. **Katrine Sjøborg:** Writing – review & editing. **Håvard O. Skjerven:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Cilla Söderhäll:** Writing – review & editing, Project administration. **Riyas Vettukattil:** Writing – review & editing, Project administration. **Magdalena R. Værnesbranden:** Writing – review & editing, Data curation. **Johanna Wiik:** Writing – review & editing, Data curation. **Anne Cathrine Staff:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Meryam Sugulle:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization.

Presentation

Part of the results were presented in an abstract and poster presentation at the 23rd World Congress of the International Society for the Study of Hypertension in Pregnancy (ISSHP) conference in Montpellier, France August 28–31, 2022.

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Declaration of competing interest

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.placenta.2025.09.006.

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